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# COMPUTERIZED EXPOSURE AND IN VIVO EXPOSURE TREATMENTS OF SPIDER FEAR IN CHILDREN: TWO CASE REPORTS

IRIS NELISSEN, PETER MURIS and HARALD MERCKELBACH

University of Limburg

**Summary** — Two spider phobic children were first given a computerized exposure treatment, and then received exposure in vivo. The cases provided no evidence for the effectiveness of the computerized exposure treatment. Exposure in vivo was found to be very successful, resulting in substantial reduction of self-reported spider fear and clear improvement on the behavioral avoidance test.

Fear of the dark, animals, blood, heights and so forth are common in childhood (see e.g., King, Hamilton & Ollendick, 1988), but apparently most childhood fears disappear with age and are "outgrown" in the normal course of development (Bauer, 1976; Ferrari, 1986). Yet, in some children, the fear becomes severe and invalidating, interfering with normal functioning. In these cases, the diagnosis of specific phobia is made (e.g., Bernstein & Borchardt, 1991; Strauss & Last, 1993). As with "subclinical" childhood fears, specific phobias in children often remit without treatment (e.g., Agras, Chapin & Oliveau, 1972; Hampe, Noble, Miller & Barrett, 1973). Nevertheless, effective treatment methods for specific phobias in children are necessary for two reasons. First, a subset of children with specific phobias have persistent symptoms, and some specific phobias continue from childhood into the adult years. Most adult animal phobics report that the fear began during early childhood years (e.g., Öst, 1987; Merckelbach, Arrindell, Arntz & de Jong, 1992). Second, specific phobias can be a serious concern when their severity or duration undermines normal development (Kendall et al., 1991). Studies have shown the negative impact of anxiety on education (e.g., Dweck & Wortman, 1982) and social adjustment (e.g., Strauss, Lease, Kazdin, Dulcan & Last, 1989).

It is a well-established fact that behavioral treatment, and especially exposure in vivo, is very effective for specific phobias in adults (e.g., Emmelkamp, 1982). Öst (1989) demonstrated that even one session of exposure in vivo results in 90% of specific phobia patients showing significant improvement.

Interestingly, Whitby and Allcock (1994) developed a computerized exposure treatment. This treatment has been applied to spider phobia — one of the most prominent fears in adults (Marks, 1987) and children (Ollendick, 1983). It consists of hierarchically structured confrontation with spiders presented on a computer screen. Because it resembles a computer play, the computerized exposure treatment might be especially interesting for those therapists who treat children with phobias. However, to our knowledge, no study has examined the efficacy of this treatment. This article describes the treatment of two spider phobic children. In both cases, the child was first given the computerized exposure treatment, and then received a one-session

Requests for reprints should be addressed to Dr Peter Muris, Department of Experimental Abnormal Psychology, Limburg University, P.O. Box 616, 6200 MD Maastricht, The Netherlands.

exposure in vivo therapy. Treatment outcome was evaluated with self-report and behavioral measures of spider fear.

## Method

### *Subjects*

Both subjects were girls who met the DSM-IV criteria (American Psychiatric Association, 1994) for specific phobia (animal type). Girl 1 (10 years) had been fearful of spiders since the age of 3. Her fear gradually increased and eventually became invalidating: she avoided gardens and wanted to have toilets and bathrooms extensively checked for spiders. She was plagued by recurrent nightmares about spiders. Her answers on the Phobic Origin Questionnaire (POQ; Öst & Hugdahl, 1981) revealed several conditioning experiences (e.g., being teased with spiders), but modeling (e.g., her schoolfriends were afraid of spiders) and negative information (e.g., having seen aversive pictures about spiders in a book) also played a role in the etiology of her phobia.

Girl 2 (9 years) had feared spiders since the age of 5. When she saw a spider, she panicked and did dangerous things such as jumping from stairs. She refused to go to bed unless her bedroom was free of spiders. The POQ indicated that her phobia started with a conditioning incident: when she was 5, she was in the bedroom and discovered a large spider crawling on the floor. She felt terrified and helpless. After several hours, her father came home and removed the spider. Her mother had received treatment for spider phobia, so it is likely that modeling also contributed to the development of her fear.

### *Assessment*

The children first underwent a computerized treatment and then received exposure in vivo therapy. At three points of time (i.e., before treatment, after computerized exposure treatment, and after exposure in vivo), they completed the *Self Assessment Manikin* (SAM; Hodes, Cook & Lang, 1985) and carried out a *Behavioral Avoidance Test* (BAT). In addition, children filled in the *Spider Phobia Questionnaire for Children* (SPQ-C; Klorman, Weerts, Hastings, Melamed & Lang, 1974; Kindt, 1994) before treatment and after the exposure in vivo.

The SAM was used as a nonverbal self-report measure of emotion. The children were asked to indicate on the SAM how they felt (affect: 1 = very positive; 9 = very negative), how anxious they were (anxiety: 1 = not at all anxious; 9 = extremely anxious), and how powerful they were (power; 1 = extremely powerful; 9 = not at all powerful) when confronted with a spider. A total SAM score was calculated, ranging from 3 to 27 with higher scores reflecting higher levels of spider fear. The SPQ-C is a reliable 29-item true/false questionnaire that measures fear of spiders. SPQ-C scores range from 0 (not at all fearful of spiders) to 29 (extremely fearful of spiders). The BAT was used to assess actual avoidance of spiders. During this test, the children were seated behind a large table. A movable glass jar containing a (medium sized) live spider was placed on the far end of the table, 3 m from the children. Children held a string connected to the glass jar and were instructed to pull the jar as near as they could tolerate. BAT performance was coded on an 11-point scale, ranging from 0, "spider still at 3 m distance", to 10, "spider on the hand".

### *Treatment*

Before treatment, the children received a short and simple explanation of the principles of exposure treatment. Further, they were assured that they had complete control, and that therapy could be stopped at any time.

The computerized exposure treatment consists of hierarchically structured confrontation with spiders which are presented on a computer screen. The program that was used in the present study contained three types of spiders: a harmless looking cartoon spider ("smiley"), a spider that goes up and down a string ("incey"), and a black housespider ("house"). The spiders could be changed with regard to size (small, medium, large, and huge) and movement (static, controlled, and free). The hierarchy started with the small, static "smiley" spider, and eventually ended with the huge, free "house" spider. Computerized exposure treatment lasted for 1 hour.

Exposure in vivo was conducted following the procedure described by Öst (1989). This treatment consists of one-session (of about 2 hrs) of hierarchically structured confrontation with real spiders in combination with modeling by the therapist.

Computerized exposure treatment and exposure in vivo were given by the same therapist. The therapist was not involved in the assessment procedures.

## Results

Scores on therapy outcome measures are presented in Table 1. In both children, self-reported spider fear as indexed by the SAM remained high after the computerized exposure treatment. Only after exposure in vivo, SAM scores decreased substantially. A similar pattern was found for the BAT: BAT performance hardly improved after computerized exposure. Yet, after exposure in vivo, both children reached the maximal BAT score of 10, i.e., "let the spider walk on your hand". The decline of SPQ scores confirmed that, on the whole, exposure in vivo treatment was effective.

## Discussion

The present cases provided no evidence for the effectiveness of this particular computerized exposure treatment of spider phobia in children. In contrast, exposure in vivo was very successful and resulted in substantial reduction of self-reported spider fear, and clear improvement on the behavioral avoidance test. The question arises why exposure in vivo was effective whereas computerized exposure was not. Foa and Kozak (1986) have proposed a model that explicates how exposure in vivo results in fear reduction. The model is based on the assumption that phobics have a memory network that contains fear-provoking elements. An example of such an element in a spider phobic child could be: "The spider will attack me". During exposure in vivo, the child is confronted with a real spider and comes to realize that the animal does not attack. In terms of Foa and Kozak's model, information discordant to the irrational elements is made available and is incorporated, resulting in a correction of the fear network. Apparently, computer presented spiders are too degraded to initiate such a corrective process. Children might consider the computer spiders to be harmless precisely because they are not real. Imaginary spiders are effective stimuli in systematic desensitization because they are perceived as real.

Relevant to the foregoing are the types of spiders that Whitby and Allcock (1994) chose for

**Table 1**

*Scores of Both Children on Therapy Outcome Measures*

| | Before treatment | After computerized exposure | After exposure in vivo |
|---|---|---|---|
| Child 1 | | | |
| Behavioral Avoidance Test | 1 | 1 | 10 |
| Self Assessment Manikin | 17 | 16 | 5 |
| Spider Phobia Questionnaire | 17 | — | 6 |
| Child 2 | | | |
| Behavioral Avoidance Test | 1 | 2 | 10 |
| Self Assessment Manikin | 20 | 18 | 8 |
| Spider Phobia Questionnaire | 16 | — | 5 |

their computerized exposure treatment. In both children, only the "house" spider elicited some fear. Assuming that provocation of fear responses is a prerequisite for any exposure treatment, the creators should have included more lifelike spiders. As one of the children remarked, the computerized exposure treatment can best be regarded as "a warming-up game before the real therapy starts".

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